Speech-Eval

Contrastive Speech Delivery Analytics with Temporal Flaw Grounding

Multimodal AI Hackathon 2026 - Track C | October 2026

Author pending confirmation; with Claude and Codex. Genuine user takes and independent ratings pending.

1. Problem and approach

Speech-Eval asks what delivery pattern differs, where it happens and what measured evidence supports the flag. It compares deliveries and reports time intervals, quoted words and numeric template explanations. Content quality is out of scope. Controlled edits supply reproducible operation labels, not independent judgments of good speaking. A uses same-text train readers; B uses the speaker's earlier take; C-lite uses sentence norms without a same-text reference. Key-moment messages are labelled heuristics, not scored judgments.

2. Contrastive dataset construction

Sources: 15 Gettysburg and 13 O Captain LibriVox readings, plus one JFK Rice and one Reagan Challenger recording. USA public-domain evidence is recorded in the release manifests and dataset card; automatic evidence checks are not human legal review. Raw hashes remain immutable.

Canonical/as-spoken differences were checked with local CTC candidate likelihoods calibrated on 300 controls, empirical fake acceptance ≤2%. This in-sample proxy is not listening or guaranteed accuracy on real disagreements. Unsettled words and dependent evidence remain masked. Number expansion retains index mappings. Acoustic audit.

MFA/WhisperX were compared on 37 words / 74 edges after 23 unclear-word exclusions. **Claude placed visual boundaries without listening**; independence from model templates was asserted, not independently established. Human spot-check is pending. Pooled median differences: MFA 52.0 ms, WhisperX 72.0 ms; MFA is primary. Both passed independent CPU rerun identity. These are differences from a limited visual reference, not established human accuracy. Bake-off.

Eligible sources receive seven effects at five edit strengths plus mixed variants, with seeded placement. One clipped O Captain source is audit-only. The release has 1,103 indexed variants / 36.26268833333334 hours: 30 originals and 1,073 edited recordings. Raw downloads/duplicate rates are not counted again. Train/dev/test have 18 / 4 / 7 speakers. Measured counts.

Type	Controlled operation	Increasing edit strength
rushing / dragging	Pitch-preserving stretch	Raise / lower speed factor
monotone	PSOLA around median pitch	Retain less pitch range
too_quiet	Flat/trailing gain	More attenuation
awkward_pause	Phrase-internal room tone	Longer inserted gap
missing_pause	Shorten existing gap	Lower retained fraction
mumbling	Zero-phase low-pass plus gain	Lower cutoff/gain

Parameters are in configs/flaws.yaml. Time maps transport source indices into the flawed timeline; labels bind type, edit-strength severity and interval. Real MFA is run separately for detection. Oracle alignments serve label/QA only; edit boundaries are not guaranteed perceptual boundaries. Genuine takes are pending.

3. Features and speaker normalisation

Praat pitch/intensity/HNR and spectral/MFCC tracks feed word, short-window and sentence aggregates. Speaker-relative transformations reduce voice/microphone offsets, not all differences:

$st(t) = 12 \log_2[f_0(t) / \text{median } f_0]$; $Lrel(t) = L(t) - \text{median } L$.

Medians use voiced frames inside aligned words. Rate (syllables/second) is not speaker-normalised. Clarity is a high-band energy ratio, not intelligibility. Comparison rate/pitch-variation/pause features are log-transformed; loudness and clarity remain dB. Uncertain-word masks exclude dependent aggregates. Feature normalisation does not silently edit stored loudness.

4. Detection, grounding and explanations

Clean train readers give word/feature median mu and robust spread s = 1.4826 median(|x - mu|). Pooled train spread and shrinkage n0 = 5 give:

$\sigma^2_{i,f} = (n\, s^2_{i,f} + n_0\, \sigma^2_{pool,f}) / (n + n_0)$; $z_{j,f} = (x_{j,f} - \mu_{i,f}) / \sigma_{i,f}$.

Whole-word mapping excludes unmatched/partial correspondences. Runs within sentences become regions; pauses are boundary events. Directional feature rules and contradictory-pair suppression are explicit. Frozen selected k is **3.50 for every type**, not the draft default 2.0. Dev severity cut points are [2.0, 2.75, 3.5, 4.059, 6591.0]; matched dev Spearman is 0.1572985835046175. The unusual final cut point/weak correlation are retained, not repaired with test or user takes. Calibration.

Templates quote measured participant/reference deviations and flagged words. Exact publication example is pending ([Pending: example_explanation]): no supplied result report contains the numeric explanation allowed by the report-only rule. Measured rule explanations are **not causal proof**.

B uses the own earlier take; JFK/Reagan test originals are within-speaker references, not population training/calibration. C checks sentence rate, pitch range, final loudness drop and non-final end pause; mumbling/awkward pause are unsupported. Number/end guidelines never alter scores.

5. Scoring and validation

For first-to-last-word duration D, duration penalty is sum(v × d/D × 10); pause penalty is sum(0.4 × v). Clip 10 - penalty to [0,10], decimal half-up round, then weight rounded categories: pace/pitch 0.25 each, pauses 0.20, volume/clarity 0.15 each. These are configurable design choices, not an external standard. Severity-3 rushing over 10% gives pace 7.0. Validation.

The report gives per-type, not aggregate, monotonicity:

Type	Median severity-score Spearman	Non-increasing %	Constant series
awkward_pause	-0.8660254037844386	85.71428571428571	3
dragging	-0.7071067811865475	71.42857142857143	2
missing_pause	-0.7071067811865475	100.0	4
monotone	-0.7071067811865475	28.571428571428573	0
mumbling	-0.8944271909999157	71.42857142857143	0
rushing	-0.7071067811865475	71.42857142857143	1
too_quiet	-0.7071067811865475	85.71428571428571	3

Each type has seven complete series. Constant ranks have undefined Spearman; non-increasing alone is not sensitivity. **Human agreement is pending.** The blind pack/scripts exist but no real rater responses. The form asks overall delivery 1-5, flaw type(s) and approximate time, not five category ratings. Scripts compute system-mean-human/per-rater Spearman, mean defined pairwise human Spearman and exact flaw-set agreement with denominators; no fabricated responses or CIs.

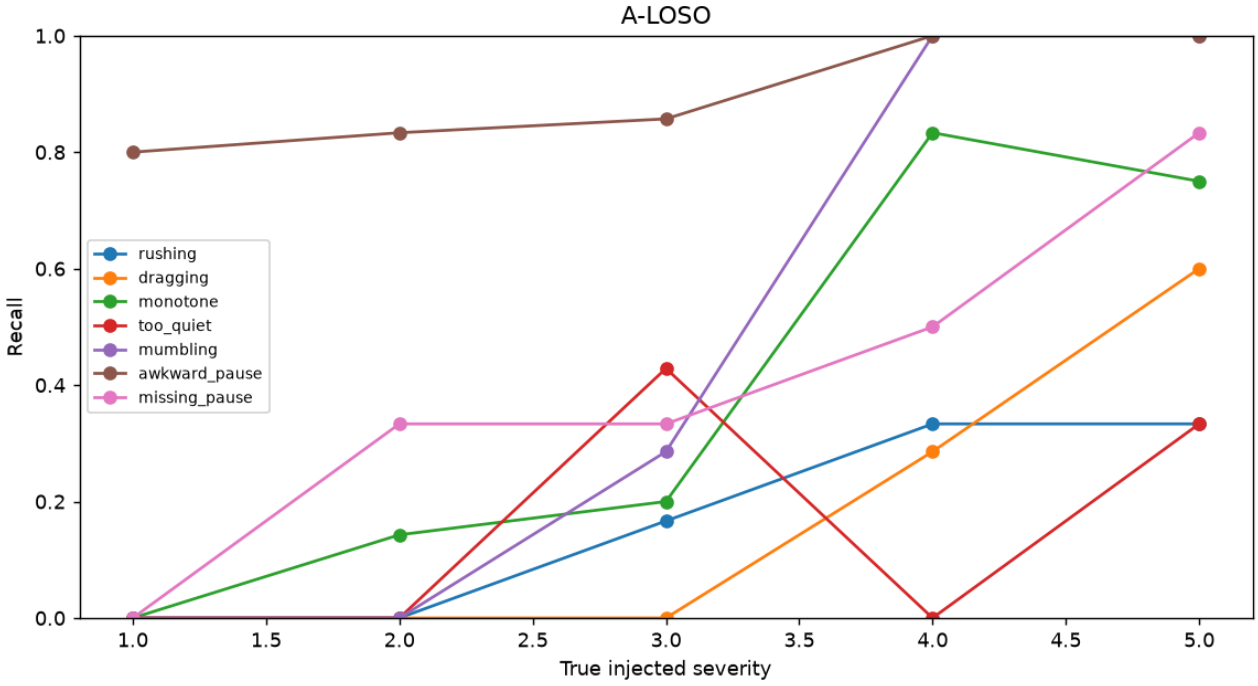
Frozen host two-run identity: 10/10. Later fresh-clone CPU Docker reproductions matched all 563 scientific outputs twice, including cross-environment checks. Scope and SHA manifests.

6. Results across settings

Thresholds used dev only. Original final test commit: aaf1832fe24a193d4c90f9e7a2ff9537b753335c. Later test runs are **reproductions**, not tuning. Same-type one-to-one region IoU ≥0.3; pause centre distance ≤0.25 s. Frozen results.

Setting	Files	Precision	Recall	F1 [95% CI]	Clean alarms/min
A-LOSO	190	0.125	0.400	0.190 [0.156, 0.223]	1.341
B-same-speaker	76	0.018	0.390	0.035 [0.026, 0.045]	0.000

A highest recall: awkward_pause 0.903; lowest: rushing 0.167. Matched overall onset/offset differences are 26.000 / 40.500 ms (A), 31.000 / 35.500 ms (B), conditional on matches, not all truth regions. Dev clean-alarm guard **failed**: 1.777 versus ≤0.5. Learned detector was not trained. No pooled severity-1/2 or severity-4/5 recall is inferred from the chart without a reported numeric summary.



A-LOSO recall versus injected severity

6. Results - no-reference comparison

C uses **different eligibility and sentence/type matching**. Its paired comparison on 175 files stays separate from Phase6:

Paired setting	Precision	Recall	Clean alarms/min
Same-text / own-original reference	0.106	0.324	0.584
C-lite norms	0.020	0.179	4.159

C too-quiet truth is trailing-only; unsupported single effects are omitted and mixed files score supported types only. Lower aggregate recall/more clean alarms describe controlled edits, not humans. Mode C. Genuine take accuracy remains pending. The real-MFA **synthetic workflow fixture** missed its injected region in A and B; no tuning followed. Phase10.

7. Limitations and future work

Edits approximate real flaws, which co-vary. Neutral reading is not ideal oratory. Few texts, microphones/speakers and English-only proxies limit generalisation. Visual reference was placed without listening; human spot-check/error estimates remain pending and unclear-word exclusion induces selection bias. Model-to-visual median is not annotation error; no 20-40 ms annotation-error claim is supported. C is a weaker norm prototype with unvalidated key-moment heuristics. Scores have constant/non-monotonic series and no human agreement yet. Low precision, failed dev guard and the fixture miss stay visible. Future work requires consented real recordings, independent labels and actual ratings; post-hoc detector changes must be labelled, not presented as frozen evaluation.

8. Reproducibility and attributions

make reproduce-quick DATA_ARCHIVE=/absolute/path/speech-eval-dataset-v1.zip imports the separately supplied archive and runs one text/three test readers. make reproduce re-verifies inputs and reproduces frozen Phase6/7/8 numbers in pinned CPU Docker, not every historical ASR/excerpt decision or human/video work. Quick: 11.829 / 11.954 seconds; full: 225.204 / 223.142 seconds after import/build. Hashes: data/release/reproduction_hashes.json and reproduction report. The narrow legacy-fixture gate exemption is a disclosed post-hoc release adapter, not a detector change; other historical inputs/code are verified.

Audio attribution: LibriVox volunteers and JFK/Reagan institutional collections; source-specific USA public-domain evidence. Archive: pending [Pending: dataset_url]; author: [Pending: author name]; video: [Pending: video_url]. Draft MIT / CC BY 4.0 grants await formal owner licence files. Remaining publication/human actions: NEXT.md.